

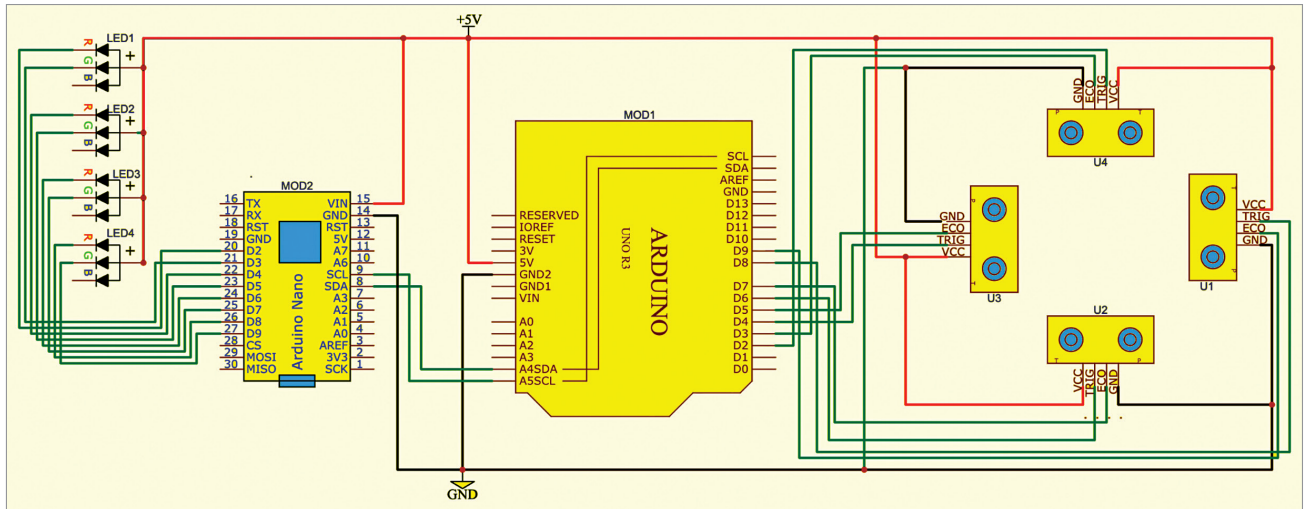


Fig. 3: Circuit diagram

TABLE 1
BILL OF MATERIALS

Component	Quantity	Description
Arduino Uno R3 (MOD1)	1	Control unit (master)
Arduino Nano (MOD2)	1	Sensor unit (slave)
HC-SR04 ultrasonic sensor (U1-U4)	4	Distance sensing for each lane
RGB LEDs (common cathode) (LED1-LED4)	4	Red and green signal simulation
Jumper wires	30+	Wiring between boards and components
Breadboard	1	For mounting and layout
USB cable (mini type B)	1 each	Programming and powering the boards

TABLE 2
CONDITION-BEHAVIOUR CORRELATION

Condition	Signal Phase	Duration	LED Behaviour
Vehicle detected	Green	10 seconds	Green on, red off
Vehicle not detected	Green	2 seconds	Green on, red off
After green (transition)	Red (all lanes)	2 seconds	Red on, green off
Other lanes (idle)	Red	Until the cycle repeats	Red on, green off

using Arduino Uno and Arduino Nano, designed to operate based on real-time vehicle detection across four lanes. HC-SR04 ultrasonic sensors detect vehicle presence and RGB LEDs simulate traffic signals. Communication between the two Arduino boards is established via the I2C protocol, ensuring coordinated sensing and signal control. The setup demonstrates intelligent lane management using low-cost and widely available hardware components.

The system is divided into two functional units: the sensor unit (Arduino Uno, master) and the

control unit (Arduino Nano, slave).

In the sensor unit, the Arduino Uno (master) continuously reads distance values from four ultrasonic sensors, each allocated to one lane. Vehicle presence is determined by comparing the measured distance against a fixed threshold of less than 15cm. The detection status for all four lanes is encoded into a single byte and transmitted to the control unit via I2C communication.

In the control unit, the Arduino Nano (slave) continuously receives detection data from the Uno and determines the traffic signal

duration for each lane. If a vehicle is detected in a lane, the Nano activates the green LED for 10 seconds. If no vehicle is detected, the system provides a brief 2-second green phase to minimise unnecessary waiting. During this time, all other lanes remain red. The process cycles through all four lanes, dynamically optimising the signal timing in response to real-time traffic flow.

This intelligent timing approach reduces idle waiting periods at intersections and enhances traffic efficiency, particularly during uneven or fluctuating traffic conditions.

Circuit diagram

The complete circuit diagram of the proposed system is shown in Fig. 3. It is built around an Arduino Uno, an Arduino Nano, four ultrasonic sensors, four common-cathode RGB LEDs, and a few other supporting components. The Arduino Uno and Nano communicate through the I2C interface using pins A4 (SDA) and A5 (SCL). The controller (Arduino Nano) receives sensor data via I2C and controls the RGB LEDs accordingly to simulate the traffic signal operation.

There are two types of RGB LEDs available in market: common cathode and common anode. Their wiring differs accordingly. For a common-cathode RGB LED, the common